

Serving Size Optimization

1 Case Setting

A high-volume Indonesian restaurant serves a variety of traditional dishes including proteins (Ayam, Rendang, Ikan) and sides (Orek Tempe, Bali Tahu). Despite steady foot traffic, the management has observed significant food waste and fluctuating profit margins. Initial audits revealed that while some dishes are polished clean, others return to the kitchen with over 50% of the portion leftover.

We build a vision model to estimate the consumption ratio for each dish and summarize daily demand and leftover data for dishes on the menu. With collected daily operational data, we employ Non-linear Programming to recalibrate serving sizes (S_i) based on observed Consumption Ratios (R_i) and Customer Sensitivity (α_i) to maximize profitability under a strict raw material budget.

2 Parameters

Symbol	Parameter	Definition / Description
i	Dish Index	Represents the specific dish ($i = 1, 2, \dots, n$).
$S_{base,i}$	Current Serving Size	The current portion size of dish i measured in grams.
R_i	Consumption Ratio	The estimated ratio ($0 \leq R_i \leq 1$) of the dish actually consumed by customers.
C_i	Marginal Cost	The raw material cost per unit of weight for dish i .
P_i	Selling Price	The fixed price at which dish i is sold to the customer.
Q_i	Current Demand	The total number of orders (units) for dish i per period.
α_i	Size Sensitivity	The coefficient measuring the elasticity of demand relative to changes in portion size.
C_{budget}	Daily Budget	The maximum allowable total budget per day.

3 Demand Response

Since the demand is decision-dependent, we model the demand decrease using a power function:

$$D_i(S_i) = D_{base,i} \cdot \left(\frac{S_i}{S_{base,i}} \right)^{\alpha_i}$$

- $\alpha = 0.1$ for side dishes(Nasi, Orek Tempe)
- $\alpha = 0.8$ for significant add-on proteins (Telur Mata Sapi, Bali Telur, Bali Tahu)
- $\alpha = 0.8$ for main dishes (Rendang Ayam, Ayam Bumbu Bistik, Ayam Bumbu Laos, Ikan Acar Kuning)

4 Non-linear Programming

Trade-off between maintaining customer satisfaction and reducing food wastes (reducing costs).

4.1 Decision Variables

We only want to optimize the serving size S_i for dishes with consumption ratio $R_i \leq 0.75$.

4.2 Objectives

The objective is to maximize total profit.

$$\text{Maximize } Z = \sum_{i=1}^n [P_i - C_i \cdot S_i] \cdot D_i(S_i)$$

4.3 Constraints

To avoid complaints, the new size must at least cover what the average person actually eats, plus a safety buffer (here we set $\epsilon = 15\%$).

$$S_i \geq S_{base,i} R_i (1 + \epsilon)$$

To reduce food waste and cost, we restricted the adjusted serving size smaller than the current size.

$$0 < S_i \leq S_{base,i}$$

Total budget constraint.

$$\sum_i (C_i \cdot S_i) \leq C_{\text{Budget}}$$